Unemployment

Primary Reference: Macroeconomics in Modules

Indraprastha Institute of Information Technology, Delhi

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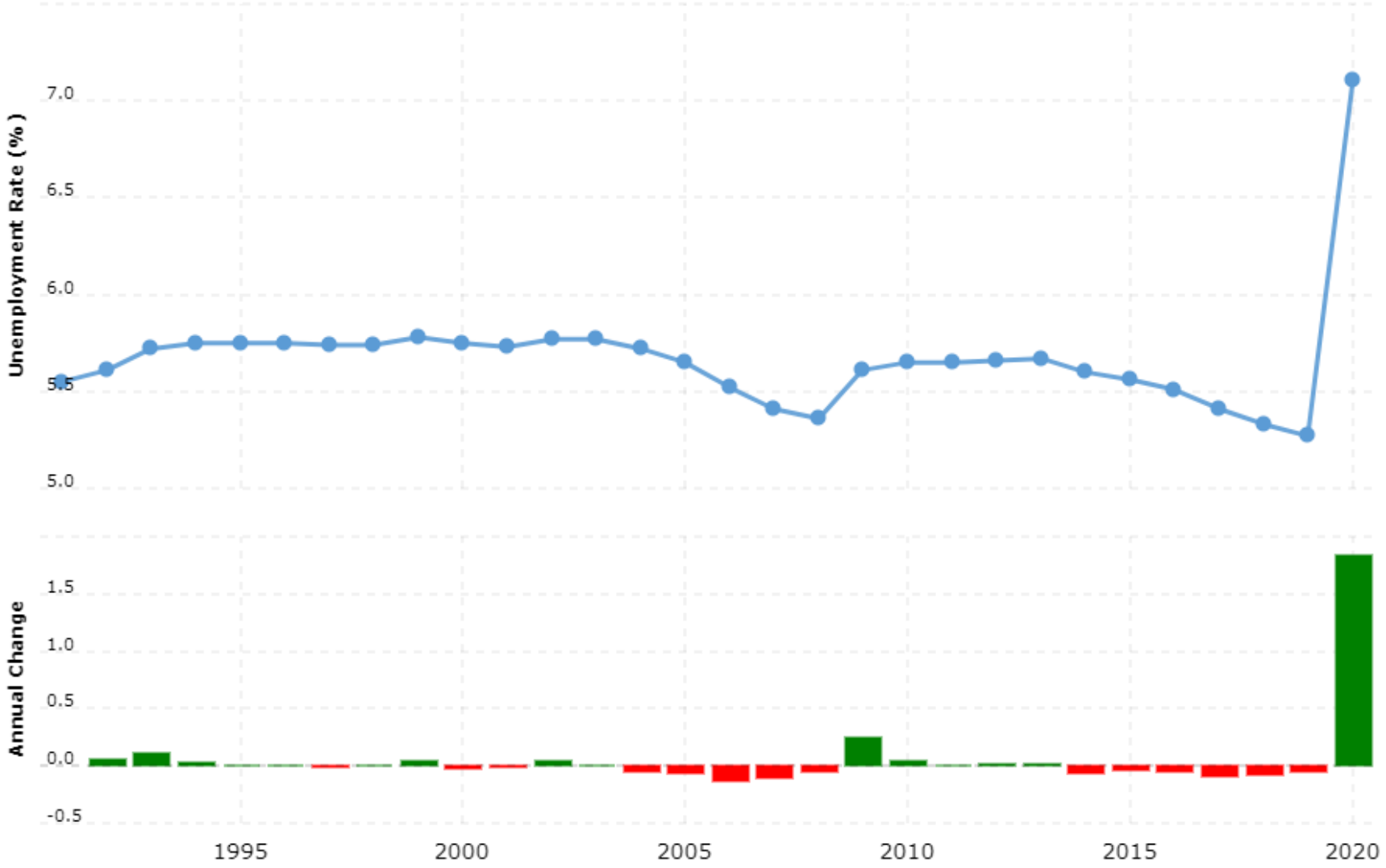
The Indian Situation



Definitions

- *Employment* is the number of people currently employed in the economy, either full time or part time.
- *Unemployment* is the number of people who are actively looking for work but are not currently employed, and are available for work.
- Labor force = Employed + Unemployed.
- The labor force consists of the population, aged 15 or older, working or looking for work.

Measurement in India

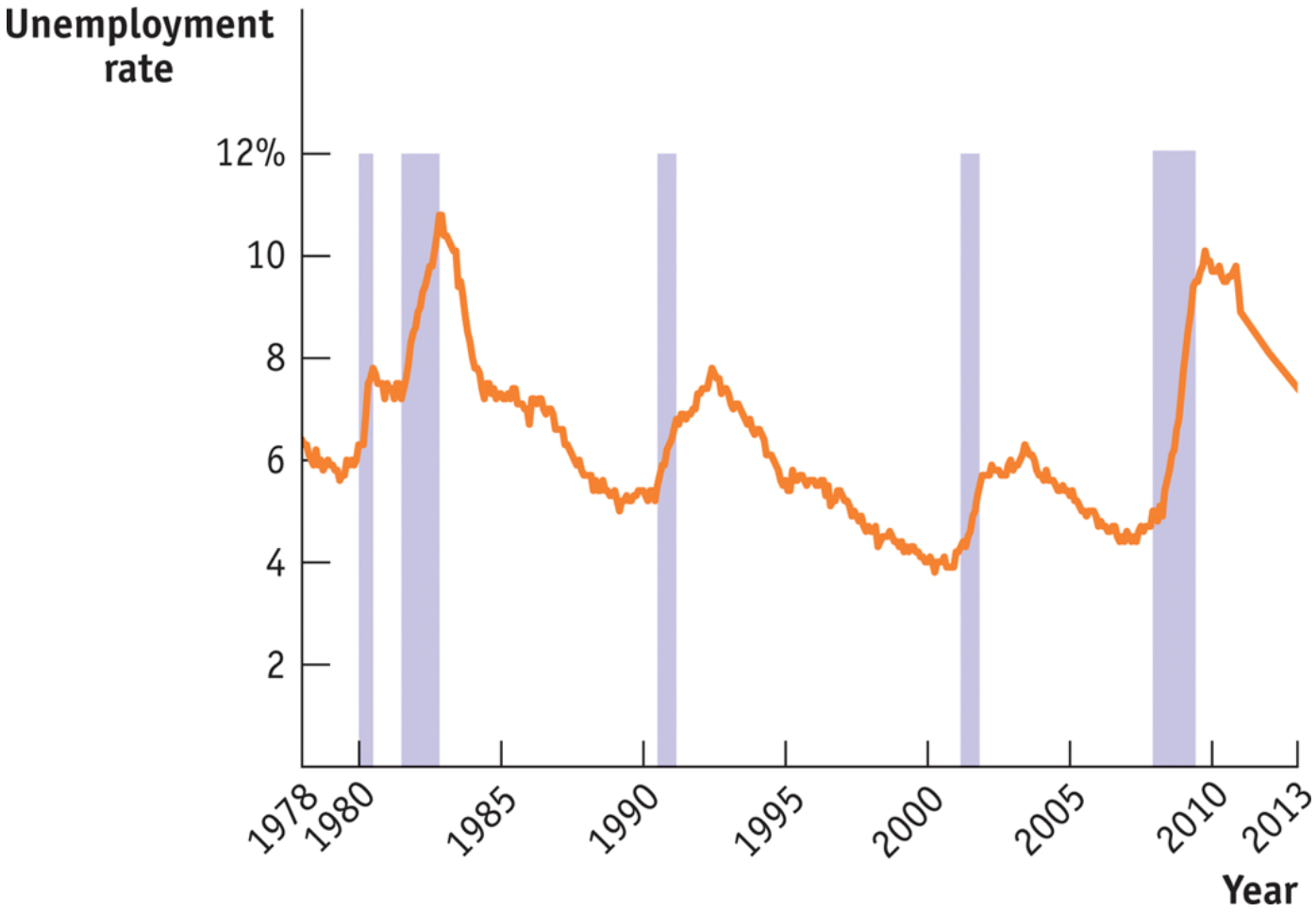
- PLFS is the survey that measures unemployment in India.
- Click here to see details.

Unemployment Rate

$$\text{Unemployment rate} = \frac{\text{Number of unemployed workers}}{\text{Labor force}} \times 100$$

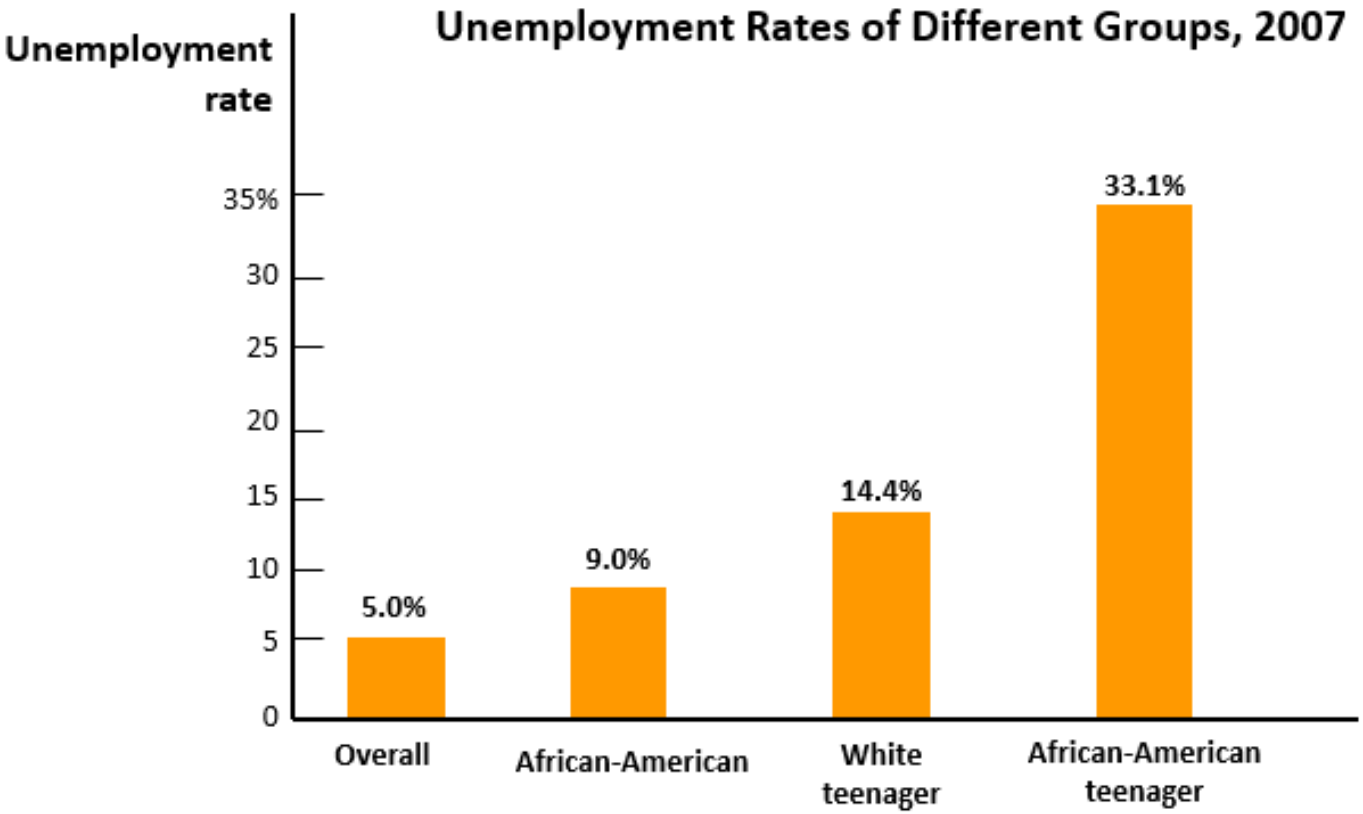
$$\text{Labor force participation rate} = \frac{\text{Labor Force}}{\text{Population age 15 and older}} \times 100$$

Unemployment and Recession



**Source: Macroeconomics in Modules by Paul Krugman and Robin Wells

Unemployment and Demographics



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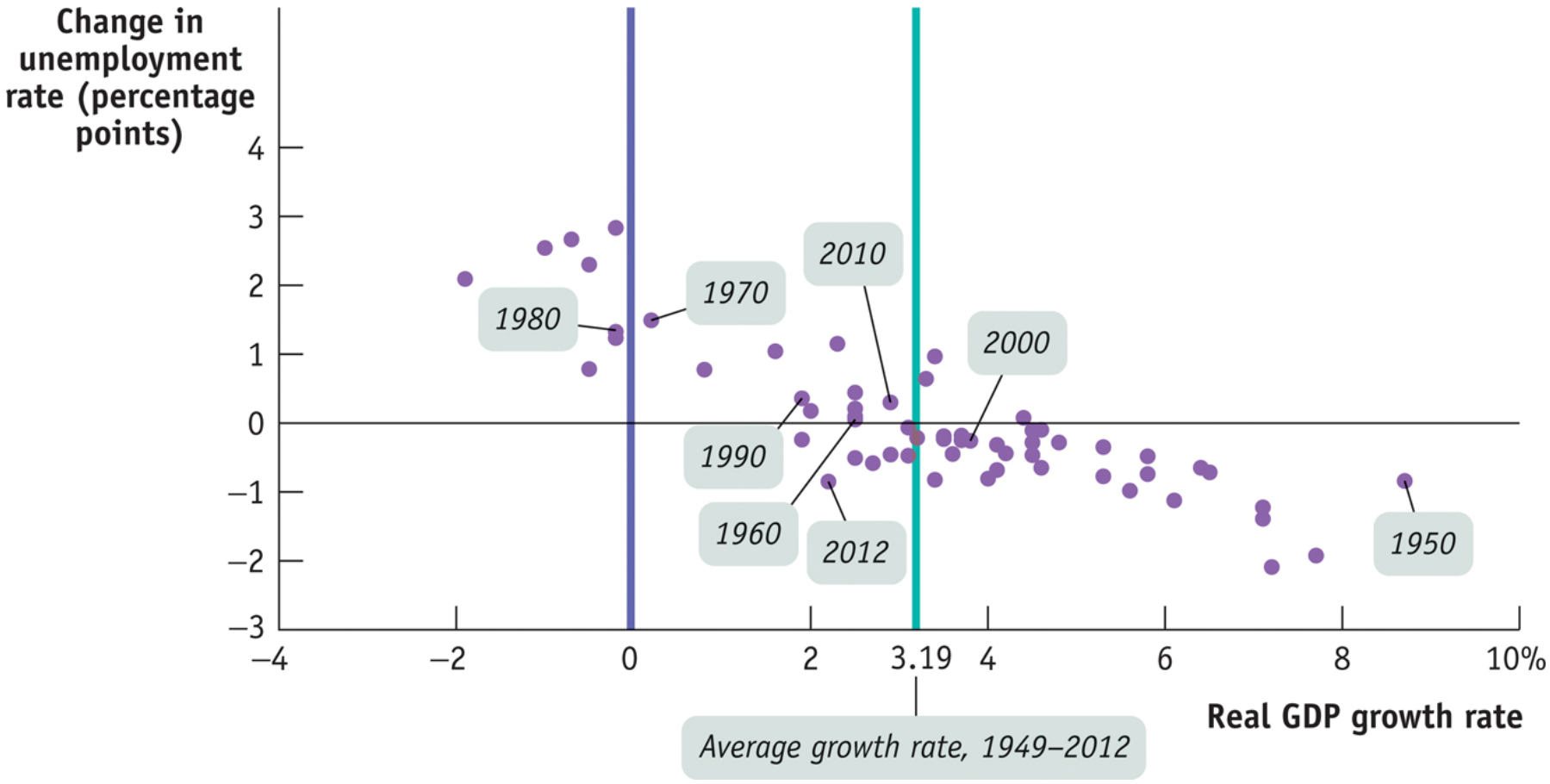
Unemployment and Growth Rate



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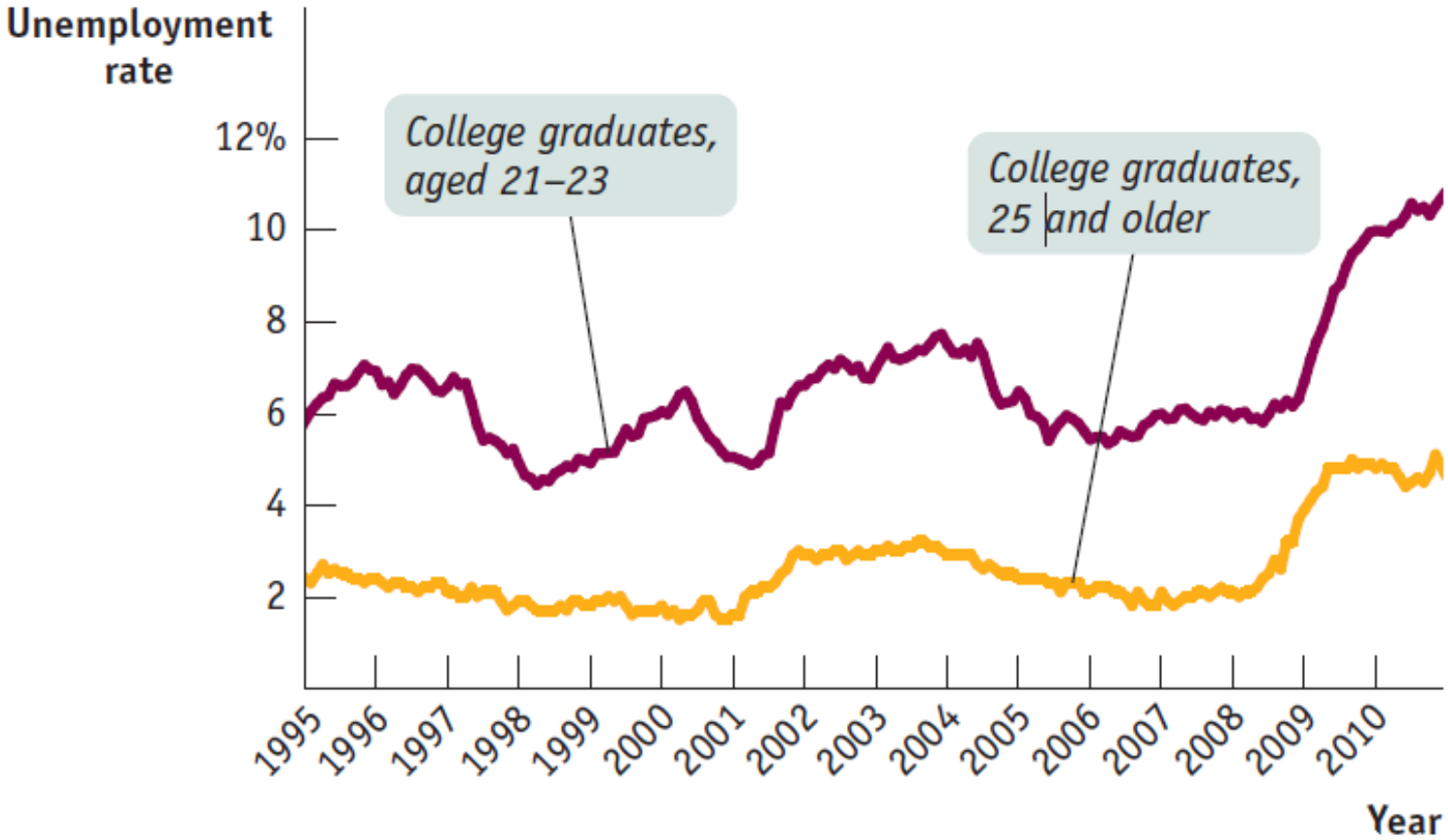
Unemployment and Age



Sources: Federal Reserve Bank of San Francisco, 2010; Bureau of Labor Statistics.

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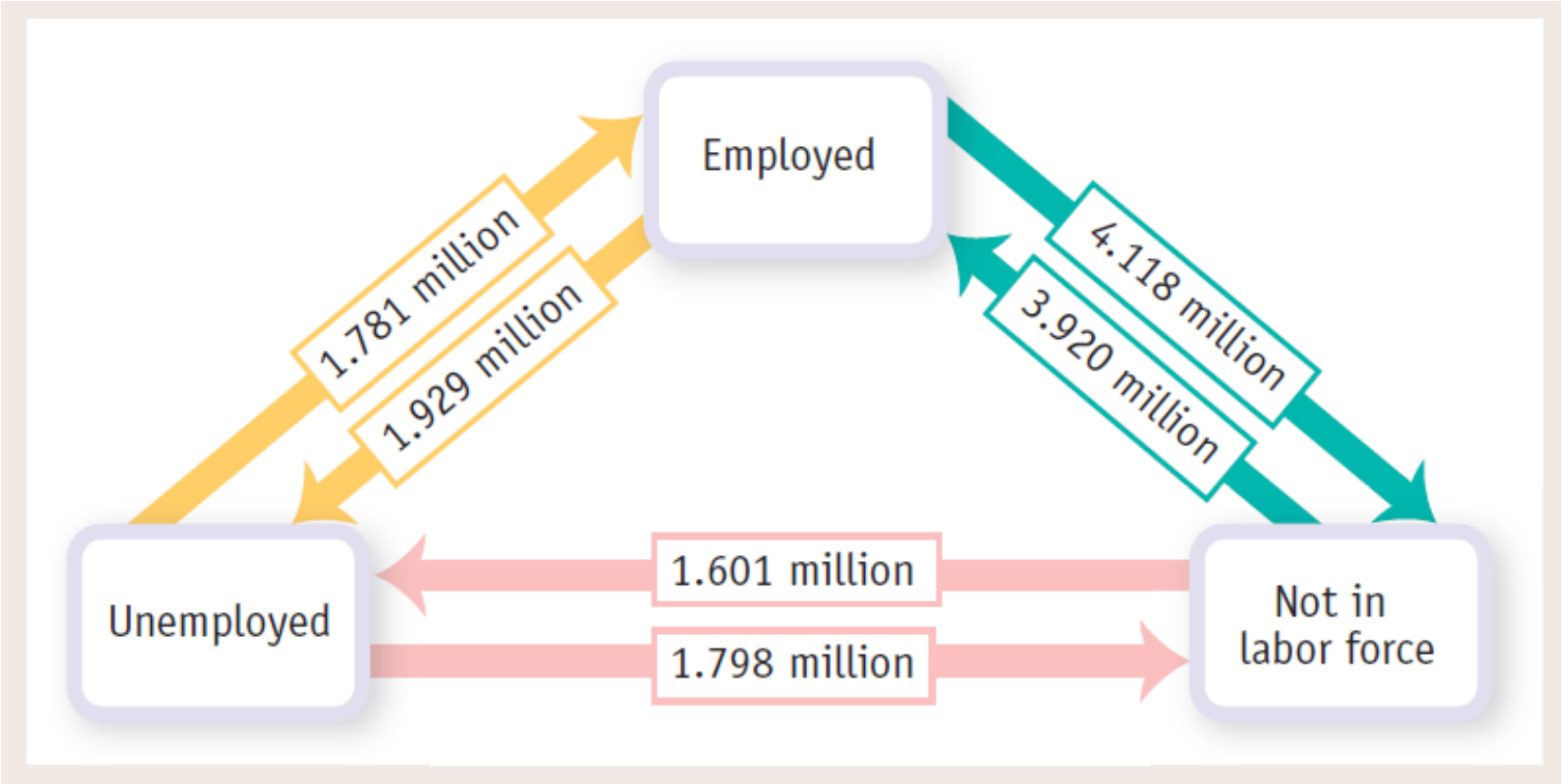
Flow of Labor Market



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In-Class Question

You will often find that the unemployment rate in USA is much higher in USA than that in India. What is the main possible reason for this?

A. The accounting is done in a different way.

B. USA is generally more strict about who it consider unemployed.

C. It's a typical problem of data mismanagement.

D. None of the above

Disguised Unemployment

- Also known as underemployment sometimes .
- People are employed on paper.
- Marginal productivity is 0 or almost 0.
- Common in small farms with many labourers.

Practical Realities

- Is 0% unemployment feasible?
- Why is that the case?

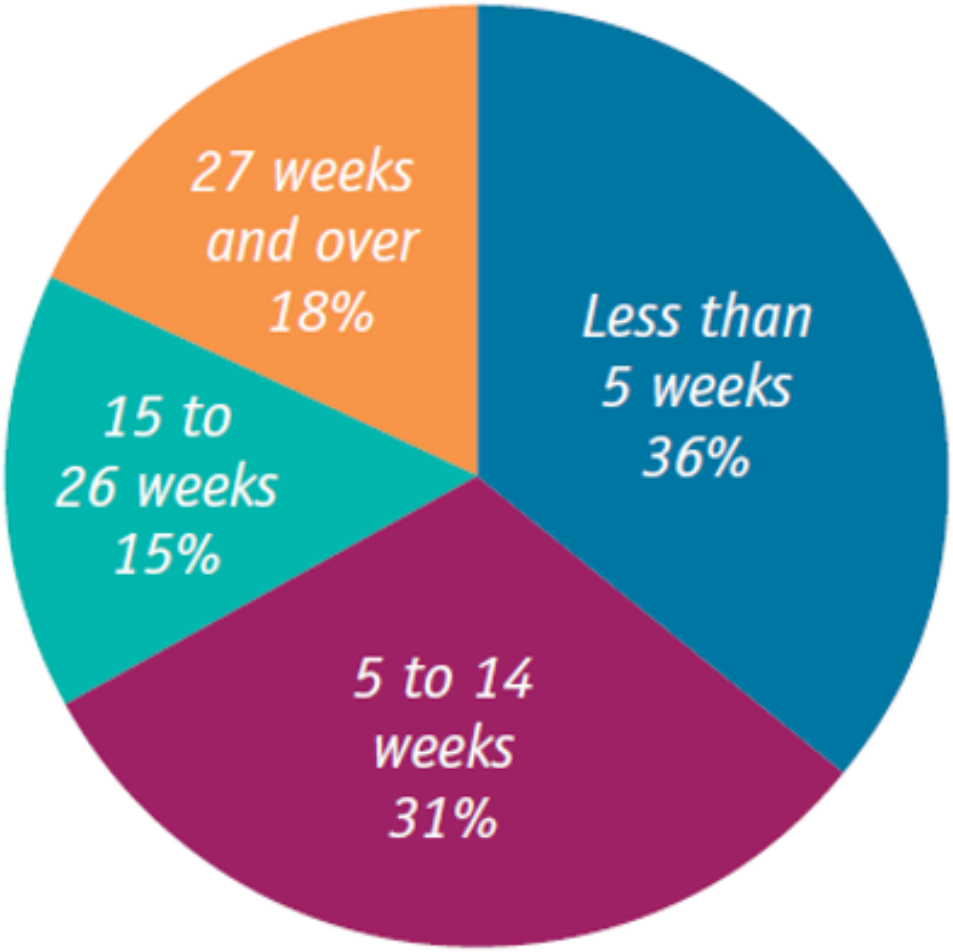
Types of Unemployment

- Frictional.
- Structural.
- Cyclical.

Frictional Unemployment

- *Frictional unemployment* is unemployment due to the time workers spend in job search.
- Part of the reason is asymmetries of information:
  - Information about jobs/ changes in the economy.
  - Information about workers.
  - Young workers.
- A limited amount of Frictional Unemployment is okay – workers and jobs take time to match skills to be efficient.
- In period of low unemployment rate, most unemployment is frictional.

Frictional Unemployment in 2007



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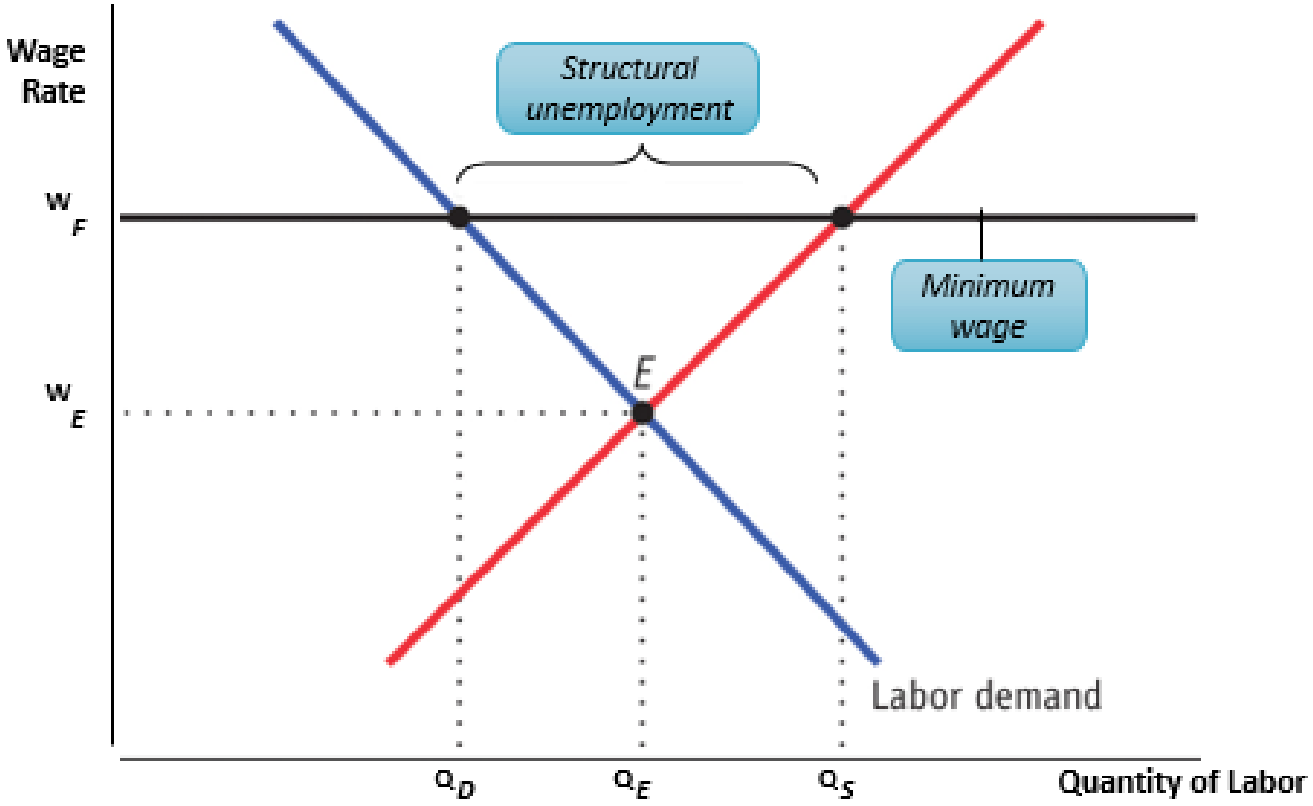
Structural Unemployment

- *Structural unemployment* is unemployment that results when there are more people seeking jobs in a labor market than there are jobs available at the current wage.
- Several reasons. Name some?

Structural Unemployment



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Structural Unemployment

- Minimum wages.
- Efficiency wages.
- Labor unions.
- Side effects of government policies.
- Mismatches between employees and employers.

Cyclical Unemployment

- Unemployment caused by the business cycle.
- Lower growth is usually correlated with higher unemployment. Why?

Cyclical Unemployment

- Unemployment caused by the business cycle.
- Lower growth is usually correlated with higher unemployment.
  - When GDP falls, firms lay off workers.
  - Idle labor and capital $\rightarrow$ economic growth not being maximized $\rightarrow$ ability of the economy to create more jobs. $\downarrow$.

The Natural and Actual Rate of Unemployment

- Even when the economy has fully recovered from a recession, so that cyclical unemployment is zero, the actual unemployment rate will still not be zero.
- Frictional and structural unemployment are always present; they are "natural."
- Natural unemployment = frictional unemployment + structural unemployment.
- Actual unemployment = natural unemployment + cyclical unemployment.

Changes in the Natural Rate

- We need estimates of the natural rate of unemployment to make forecasts - and to analyze policy.
- What causes it to change?
  - Changes in characteristics of the labor force.
    - Changing demographics.
  - Changes in labor market institutions.
    - Unions, temporary employment agencies, and new technology.
  - Changes in government policies.
    - High minimum wage $\rightarrow$ structural unemployment.
    - Unemployment benefits $\rightarrow$ structural $+$ frictional unemployment.
    - Job training and employment subsidies reduce the natural rate of unemployment.